

Выполним эквивалентное преобразование схемы и определим комплексные параметры цепи

$$\omega = 2\pi f = 2\pi \cdot 50 = 314.1593 \text{ rad/s}$$

$$E_1 = \frac{U_1}{\sqrt{2}} e^{j\omega t} = 17.678 e^{j25^\circ} \text{ V}$$

$$I_2 = \frac{I_1}{\sqrt{2}} e^{j\omega t} = 10.607 e^{j15^\circ} \text{ A}$$

$$E_2 = \frac{U_2}{\sqrt{2}} = \frac{10.25 \cdot 10^{-3}}{0.1} = 102.45 + j27.45 \text{ V}, R_{in2} = \frac{1}{\sigma_2} = \frac{1}{0.1} = 10 \Omega$$

$$E_{eq} = E_1 - E_2 = 16.02 + j7.47 - (102.45 + j27.45) = -86.43 - j19.98 \text{ V}$$

$$Z_{eqm} = (R_{in1} + R_{in2}) + j\omega L_{in1} = (5 + 10) + j2 \cdot 15 = 15 + j2 \Omega = 15.13 e^{j7.59^\circ} \Omega$$

Рассчитаем эквивалентные сопротивления ветвей приемника

$$X_{L1} = \omega L_1 = 314.1593 \cdot 0.006 = 1.885 \Omega, X_{C1} = \frac{1}{\omega C_1} = \frac{1}{314.1593 \cdot 0.00004} = 67.726 \Omega$$

$$X_{L2} = \omega L_2 = 314.1593 \cdot 0.11 = 34.558 \Omega, X_{C2} = \frac{1}{\omega C_2} = \frac{1}{314.1593 \cdot 0.000035} = 90.946 \Omega$$

$$X_{L3} = \omega L_3 = 314.1593 \cdot 0.07 = 21.991 \Omega, X_{C3} = \frac{1}{\omega C_3} = \frac{1}{314.1593 \cdot 10^{-6}} = 0 \Omega$$

$$Z_1 = R_1 + jX_{L1} - jX_{C1} = 4 + j1.885 - j67.726 = 4 - j65.84$$

$$Z_2 = R_2 + jX_{L2} - jX_{C2} = 16 + j34.558 - j90.946 = 16 - j56.39$$

$$Z_3 = R_3 + jX_{L3} - jX_{C3} = 17 + j21.991 - j0 = 17 + j21.99$$

Находим эквивалентное сопротивление нагрузки и входное сопротивление цепи

$$Z_{23} = \frac{Z_2 Z_3}{Z_2 + Z_3} = \frac{(16 - j56.39)(17 + j21.99)}{33 + j34.4} = 31.15 + j14.08$$

$$Z_{eqload} = Z_1 + Z_{23} = 4 - j65.84 + (31.15 + j14.08) = 35.15 - j51.76$$

$$Z_m = Z_{eqm} + Z_{eqload} = 15 - j2 + (35.15 - j51.76) = 50.15 - j49.76$$

Определяем значения токов

$$I_1 = \frac{E}{Z} = \frac{-86.43 - j19.98}{50.15 - j49.76} = -0.67 - j1.06 \text{ A}$$

$$I_2 = I_1 \frac{Z_3}{Z_2 + Z_3} = -0.67 - j1.06 \cdot \frac{17 + j21.99}{33 + j34.4} = 0.67 - j0.29 \text{ A}$$

$$I_3 = I_1 - I_2 = -0.67 - j1.06 - (0.67 - j0.29) = -1.34 - j0.77 \text{ A}$$

Выполняем проверку расчетов по законам Кирхгофа

$$\Delta I = 0 - j0, \Delta U_1 = 0 + j0, \Delta U_2 = -0 + j0$$

$$i_{KCL} = 0^0_0, i_{KVL} = 0^0_0$$